Academic Session 2025-26

EVEN Semester Jan-Jun 2026

UNIVERSITY INSTITUTE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BTECH CSE – CS 201

SEMESTER: 6TH

CLOUD COMPUTING

(23CSH-307)

Unit No. 1 **Chapter No. 1** **Lecture No. 1**

Topic : Introduction to Cloud Computing

Sanjay Tiwari, E18823 **Senior Technical Trainer**

Learning Objectives

- Understand what cloud computing fundamentally means
- Learn how computing evolved from mainframes to the cloud era
- Identify generic, universally accepted characteristics of cloud computing
- Distinguish cloud as a modern computing paradigm
- Recognize the historical motivations behind cloud adoption
- Understand how abstraction influenced cloud development
- Build conceptual clarity necessary for advanced cloud topics
- Stay focused on the three allowed content areas only

Definition of Cloud Computing

- Cloud computing provides IT resources over the internet
- Resources include computing power, storage, platforms, and tools
- Users access services without installing or maintaining hardware
- Computing resources are provided from large remote data centers
- Users can request resources whenever needed
- Providers automate delivery of computing services
- Consumers pay based on usage or subscription
- Cloud abstracts infrastructure complexity from end users

What Makes Something “Cloud Computing”?

- Services delivered remotely through the internet
- Resources available instantly without manual setup
- Systems designed to scale up and down efficiently
- Infrastructure shared among many users
- Remote access from multiple device types
- Automated management by the provider
- Flexible subscription or consumption-based model
- High availability through distributed infrastructure

Evolution of Computing: Mainframes

- **1950s–1970s:** Computing started with powerful centralized machines
- Users connected through basic terminals with no processing power
- Data and applications resided entirely on the central mainframe
- Systems were secure, robust, but extremely expensive
- Limited flexibility due to tightly controlled environments
- Enterprises depended heavily on vendor-specific architectures
- Resource sharing was primitive and restricted
- This era established the idea of centralized service delivery

Evolution: Client–Server Era

- Emerged in **1980s–1990s** with personal computers
- Workload distributed between clients and local servers
- Applications ran on organizational servers rather than mainframes
- Enabled more flexibility compared to centralized systems
- Organizations had to purchase, install, and maintain hardware
- Scaling required physically adding new servers
- Infrastructure costs increased as user demands grew
- Set the stage for exploring more efficient computing paradigms

Evolution: Virtualization & Distributed Systems

- Early 2000s brought virtualization technologies
- Allowed multiple computing environments on one physical server
- Improved hardware utilization significantly
- Reduced cost of running large computing systems
- Distributed systems enabled workloads to run across machines
- Internet infrastructure improved globally
- Software automation became capable of managing complex systems
- These advancements collectively led to early cloud foundations

Evolution: Cloud Era Emerges

- Mid-2000s: providers began offering computing via the internet
- Large data centers operated at global scale
- Users requested servers, storage, or apps remotely
- Instant provisioning replaced slow hardware setup
- Resource flexibility matched modern application demands
- Online platforms offered standardized computing environments
- Enterprises moved away from local infrastructure dependency
- Cloud became a dominant IT delivery model worldwide

Core Characteristics of Cloud Computing

- **Instant Availability:** Computing resources delivered immediately
- **Remote Accessibility:** Services accessed from any internet-connected device
- **Scalability:** Ability to expand or reduce resources easily
- **Shared Infrastructure:** Multiple users share large resource pools
- **Automation-Driven:** Systems allocate resources automatically
- **Resilience:** Infrastructure designed to remain operational despite failures
- **Usage-Based Billing:** Users pay proportionally to what they consume
- **Service Abstraction:** Hardware complexities hidden from the user

Why These Characteristics Matter

- Together, they define cloud as a modern computing approach
- They ensure flexibility for diverse application workloads
- They reduce dependency on physical hardware ownership
- They enable rapid development and deployment cycles
- They support global application accessibility
- They allow computing to be treated as a utility
- They form the basis of modern IT management practices
- They distinguish cloud from earlier computing eras

Summary

- Cloud computing is remote, automated, internet-based IT service delivery
- It abstracts physical infrastructure from the user
- It evolved through mainframes, client–server, virtualization, and distributed systems
- Cloud emerged due to need for scalability and flexibility
- Providers built massive data centers enabling global services
- Characteristics include scalability, automation, shared resources
- Billing models align with consumption rather than ownership
- These fundamentals shape all modern cloud technologies

Next Lecture

Comparison of cloud computing with traditional IT infrastructure, key benefits of cloud adoption, limitations and challenges

FAQ'S

- Was cloud computing possible during mainframe era?
  No, network and virtualization technologies were not mature.
- Why is cloud considered a new computing paradigm?
  Because it abstracts hardware and delivers computing as a service.
- Do users need to maintain hardware in cloud?
  No, providers manage all underlying infrastructure.
- Is cloud dependent on internet?
  Yes, internet is required for service delivery.
- Why was client–server model insufficient?
  It could not scale as rapidly as modern workloads required.

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Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

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- Cloud Computing: https://onlinecourses.nptel.ac.in/noc26_cs55/preview
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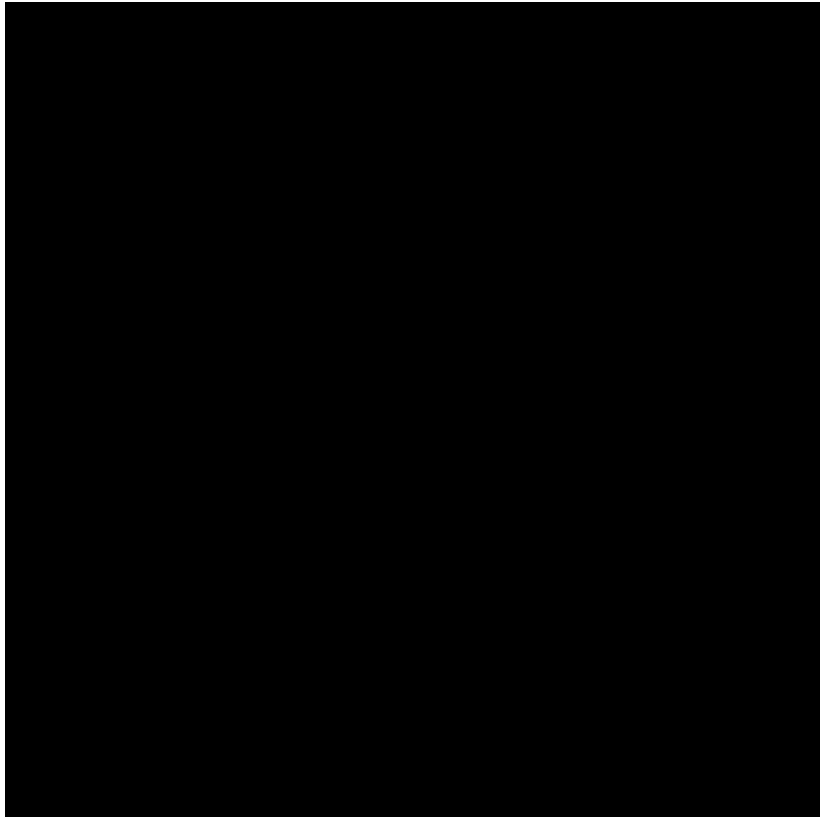

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CLOUD COMPUTING

(23CSH-307)

Unit No. 1 **Chapter No. 1** **Lecture No. 2**

Topic : Comparison of cloud computing with Traditional Infrastructure

Sanjay Tiwari, E18823 **Senior Technical Trainer**

Learning Objectives

- Compare cloud computing with traditional on-premises IT setups
- Understand differences in management, cost, flexibility, and scalability
- Learn business-level advantages of adopting cloud environments
- Identify common challenges faced during cloud transition
- Evaluate operational trade-offs between cloud and traditional IT
- Understand security, governance, and reliability considerations
- Build decision-making insight for IT modernization
- Stay within the exact topic boundaries requested

Traditional IT Infrastructure: Key Characteristics

- Organizations purchase and maintain physical hardware
- Requires dedicated data center space, cooling, and power
- Upgrades require capital investment and procurement cycles
- IT teams manage servers, storage, and networks manually
- Capacity planning must be done years in advance
- Scaling requires buying new equipment
- Hardware failures require on-site intervention
- Security and compliance fully managed by internal teams

Cloud Computing Infrastructure: Key Characteristics

- Infrastructure hosted externally by a service provider
- Users access computing resources remotely
- Resources can be adjusted without physical installation
- Costs typically shift from capital to operational expenses
- Maintenance is largely handled by the service provider
- Global access possible from multiple locations
- Providers offer managed tools for operations and monitoring
- Reduces the need for large internal data center teams

Cloud vs Traditional IT: Comparative Analysis

- Traditional IT requires upfront hardware investment; cloud does not
- Cloud allows flexible resource adjustments; traditional IT is rigid
- Traditional scaling is slow; cloud scaling is faster (without explaining elasticity)
- Cloud reduces physical space requirements; traditional IT depends on data centers
- Traditional IT involves full hardware ownership; cloud uses provider infrastructure
- Support and maintenance differ significantly between models
- Cloud simplifies global access; traditional IT is location-bound
- Traditional IT offers full control; cloud introduces shared operational responsibility (NOT the shared responsibility model — only the idea of divided duties)

Key Benefits of Cloud Adoption

- Reduced capital expenditure on hardware & data centers
- Faster deployment of applications and services
- Ability to adjust resource usage based on need
- Lower burden on internal IT teams
- Access to advanced tools without in-house expertise
- Improved accessibility from remote and distributed locations
- Providers handle routine updates and maintenance
- Enhanced reliability due to geographically distributed infrastructure

Additional Advantages

- Faster disaster recovery through provider infrastructure
- Simplified backup and data retention processes
- Better support for collaborative and remote work
- Higher operational efficiency through automation tools (generic mention)
- Reduced risks of hardware obsolescence
- Cost tracking becomes more transparent through usage metrics
- Easier experimentation for innovation projects
- Eliminates procurement delays for new IT capacity

Limitations & Challenges of Cloud Adoption

- Initial migration can be complex and time-consuming
- Data transfer between systems may require careful planning
- Organizations may face dependency on external providers
- Applications not designed for distributed environments may need redesign
- Cloud cost mismanagement can lead to overspending
- Internet connectivity becomes mission-critical
- Governance and policy alignment must be re-evaluated
- Some industries face stricter compliance and regulation challenges

Cloud Adoption Risks

- Vendor lock-in if services become difficult to migrate away from
- Misconfiguration risks due to unfamiliar tools
- Monitoring distributed workloads may require new skills
- Legacy systems may not transition easily
- Organizational culture may resist cloud-driven change
- Training needs increase for IT staff
- Cost forecasting becomes dynamic instead of fixed
- Requires clear security and access-control strategies

Summary of the Lecture

- Traditional IT relies on owned hardware and local management
- Cloud relies on provider-operated infrastructure
- Cloud offers flexibility, faster deployment, and lower maintenance burden
- Traditional IT provides full control but requires heavy investment
- Cloud adoption offers many operational and strategic benefits
- Migration introduces challenges such as complexity and cost control
- Cloud suitability varies by workload and organizational needs
- A balanced evaluation is essential before transitioning

Next Lecture

Detailed exploration of IaaS, PaaS, and SaaS with real-world examples, understanding the shared responsibility model

Quiz/ FAQ's

- Does cloud completely eliminate the need for IT staff?
- No, staff roles shift toward management and oversight.
- Is cloud always cheaper than traditional IT?
- Not always—cost depends on usage and governance.
- Why do some companies remain on traditional IT?
- Due to compliance, legacy systems, or cost predictability.
- Is internet connectivity required for cloud?
- Yes, cloud access depends on stable network connections.
- Does cloud guarantee immediate savings?
- Savings occur only with proper resource planning.

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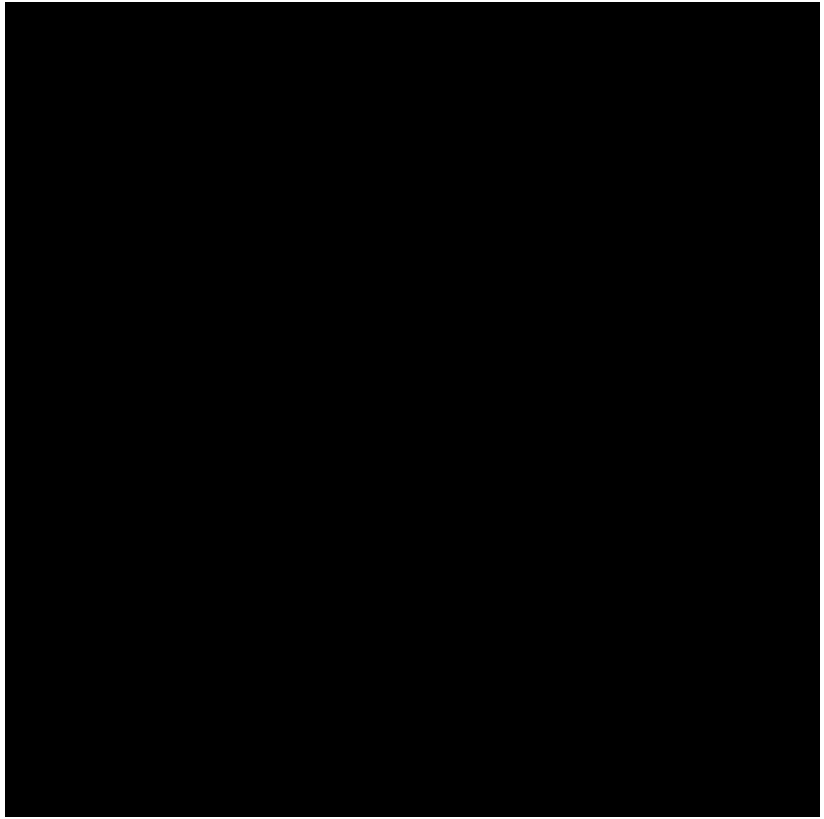

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Unit No. 1 **Chapter No. 1** **Lecture No. 3**

Topic : IaaS, PaaS and SaaS

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Learning Objectives

- Understand the purpose and structure of IaaS, PaaS, and SaaS.
- Explore real-world examples to differentiate each service model.
- Learn how control and responsibilities shift across service models.
- Understand the shared responsibility model in cloud service usage

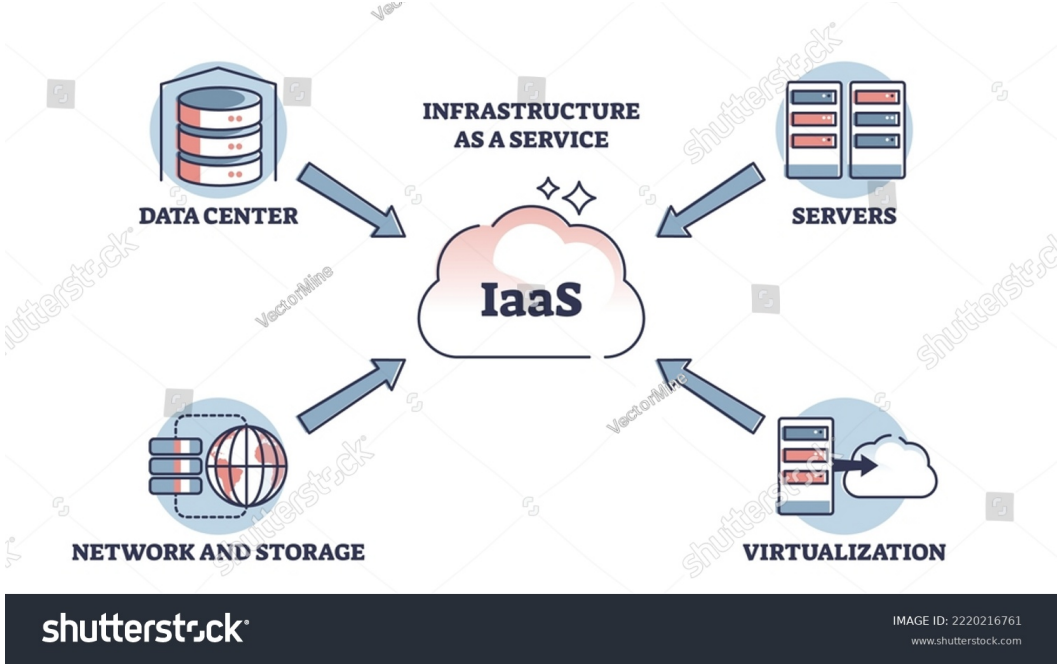
Infrastructure as a Service (IaaS): Key Concepts

- Provides virtualized computing resources like VMs, storage, and networks.
- Users manage operating systems, applications, and data.
- Offers flexible resource provisioning without physical hardware ownership.
- Providers handle data centers, physical servers, and networking hardware.
- Suited for building custom IT environments.
- Enables organizations to configure systems based on needs.
- Provides high control over configuration and security settings.
- Used commonly for hosting apps, databases, and enterprise workloads.

Real-World Examples of IaaS

- Amazon EC2 — custom virtual servers for general workloads.
- Azure Virtual Machines — Windows and Linux server deployment.
- Google Compute Engine — scalable compute instances.
- IBM Cloud Virtual Servers — enterprise-grade configurable VMs.
- Oracle Cloud Infrastructure Compute — high-performance workloads.
- DigitalOcean Droplets — developer-friendly virtual machines.
- Linode Compute — simple VM hosting for developers.
- Common uses: server hosting, testing environments, and custom deployments.

Infrastructure as a Service (IaaS): Key Concepts



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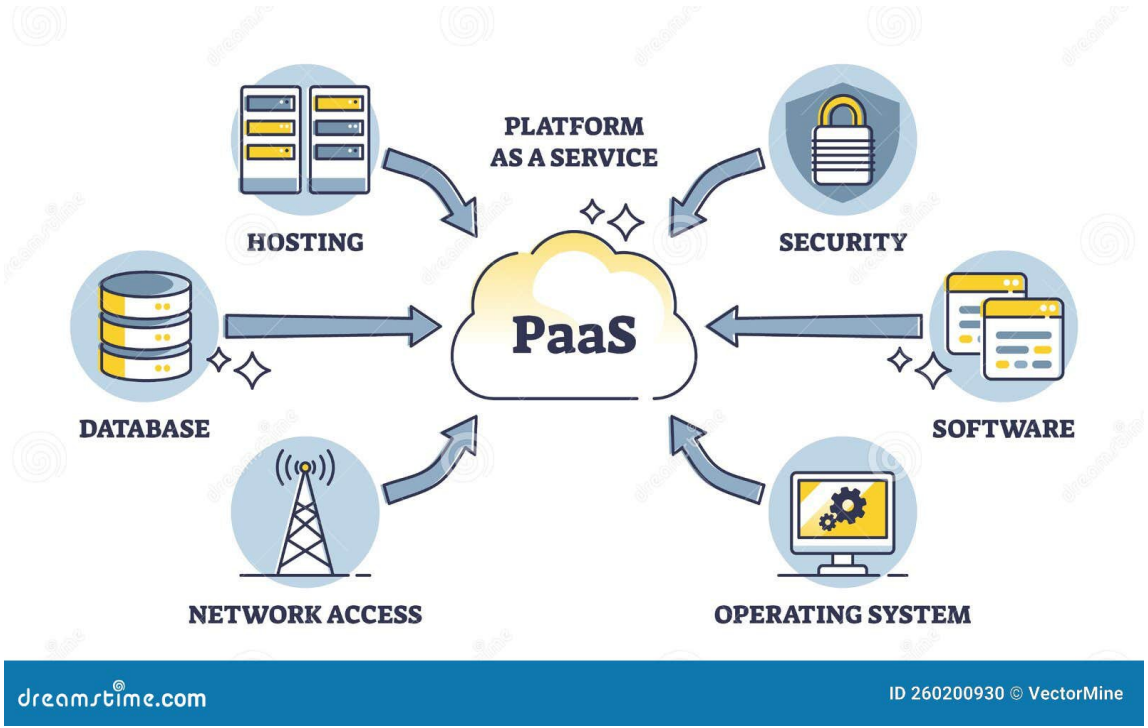
Platform as a Service (PaaS): Key Concepts

- Provides a pre-built platform with OS, runtime, and development tools.
- Developers focus on application code instead of infrastructure management.
- Providers handle updates, patches, and environment optimization.
- Eliminates server installation and manual configuration tasks.
- Speeds up application development and deployment cycles.
- Ensures consistent development environments across teams.
- Integrates CI/CD tools, monitoring, and logging.
- Ideal for modern web and mobile application development.

Real-World Examples of PaaS

- Google App Engine — deploy apps without server management.
- Microsoft Azure App Service — host web & mobile applications easily.
- AWS Elastic Beanstalk — automatic provisioning of app environments.
- Heroku — streamlined code deployment for developers.
- Render / Railway — modern managed deployment platforms.
- Supports languages like Python, Java, Node.js, Go, PHP, Ruby.
- Provides built-in scaling, monitoring, and logging tools.
- Ideal for prototypes, startups, and enterprise app builders.

Platform as a Service (PaaS): Key Concepts



Source: dreamstime

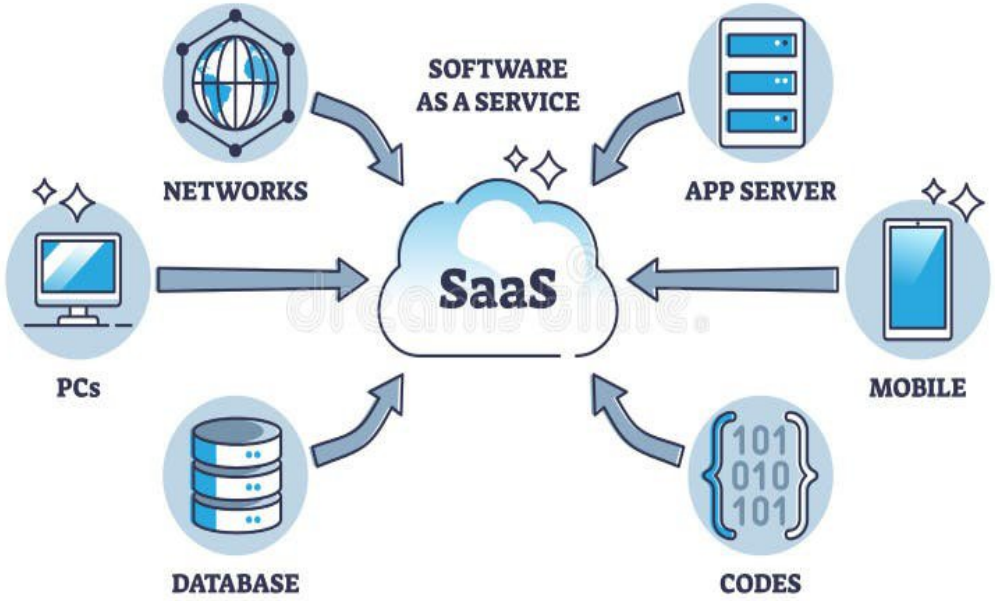
Software as a Service (SaaS): Key Concepts

- Delivers complete applications over the internet.
- Accessible through web browsers or mobile apps.
- No installation or local infrastructure required.
- Provider handles updates, maintenance, and security patches.
- Users access software using subscription or usage-based models.
- Supports multi-user access across devices and locations.
- Does not require users to manage backend systems.
- Ideal for communication, collaboration, and business tools.

Real-World Examples of SaaS

- Google Workspace — Docs, Sheets, Slides, Gmail services.
- Microsoft 365 — Word, Excel, PowerPoint online.
- Salesforce — CRM tools for sales and marketing teams.
- Zoom — cloud-based video conferencing.
- Slack — team communication and messaging.
- Dropbox — cloud-based file storage.
- Notion — all-in-one productivity and project management.
- Widely used in enterprises and educational institutions.

Software as a Service (SaaS): Key Concepts

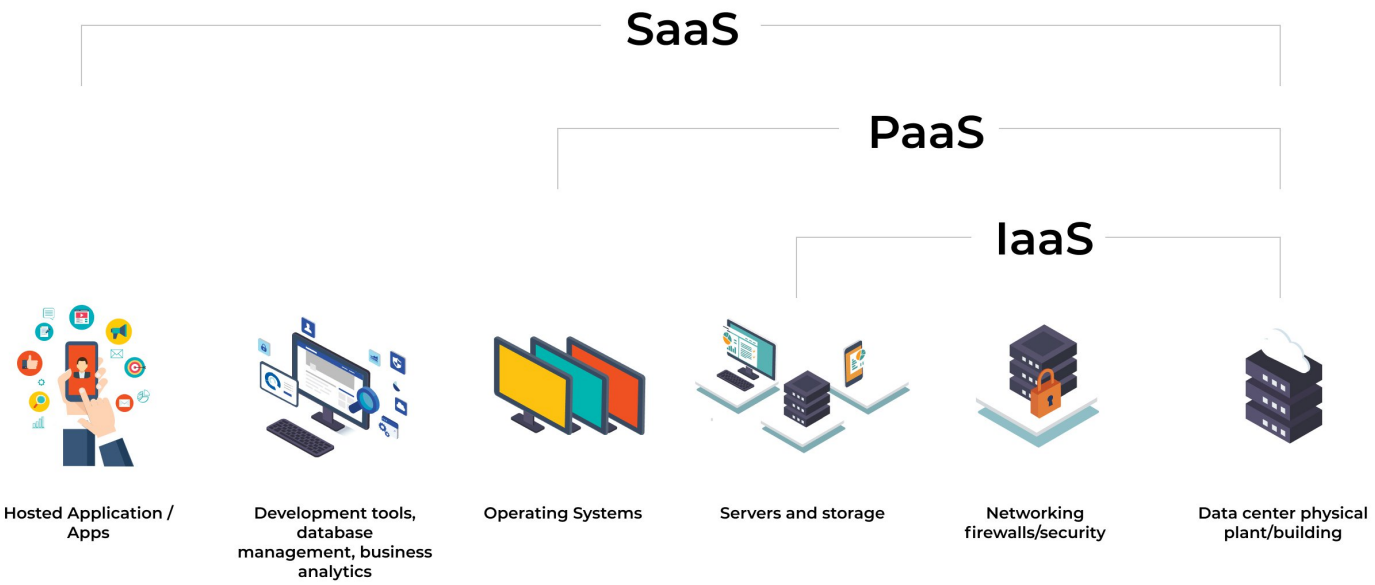


Source: dreamstime

Comparison: IaaS vs PaaS vs SaaS

- IaaS offers highest control; SaaS offers lowest control; PaaS sits in the middle.
- IaaS requires users to manage OS; PaaS eliminates OS management; SaaS eliminates app management.
- SaaS requires minimal configuration from users.
- PaaS accelerates development but limits low-level infrastructure control.
- IaaS fits customizable systems; PaaS fits developers; SaaS fits end users.
- Maintenance responsibility decreases from IaaS to SaaS.
- User involvement highest in IaaS, lowest in SaaS.
- Different service models fit different adoption needs.

Comparison



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Shared Responsibility Model

- Defines which tasks are handled by the cloud provider vs the customer.
- Provider manages physical hardware, data center security, and core systems.
- Customer manages application logic, data access, and user controls.
- In IaaS: customer manages OS, applications, and configurations.
- In PaaS: customer manages app code and data only.
- In SaaS: customer manages users and usage policies.
- Mismanagement of responsibilities can lead to vulnerabilities.
- Clear understanding is essential before using any service model.

Summary of the Lecture

- IaaS, PaaS, and SaaS offer three different levels of responsibility and control.
- IaaS provides infrastructure; PaaS provides platforms; SaaS provides applications.
- Real-world examples show how companies use these models.
- Shared responsibility differs across the three models.
- Proper understanding prevents misconfiguration and security gaps.
- Service selection depends on development, control, and operational needs.
- SaaS simplifies usage; IaaS provides flexibility; PaaS speeds development.
- These models form the foundation of modern cloud solutions.

Quiz/ FAQ's

- Can a company use all three service models together?
- Does SaaS require installation on computers?
- Why do developers choose PaaS?
- Does IaaS replace physical servers for customers?

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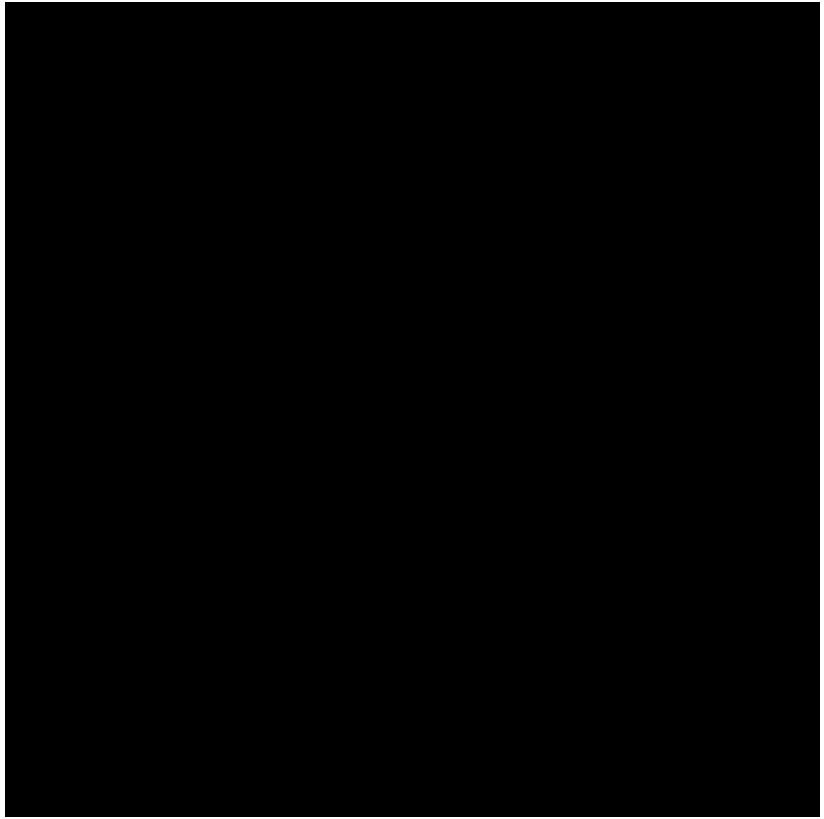
Next Lecture

Public cloud, private cloud, hybrid cloud, and multi-cloud architectures, decision criteria for deployment selection

Class-Wise Feedback



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Unit No. 1 **Chapter No. 1** **Lecture No. 4**

Topic : Public, Private and Hybrid Cloud

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Learning Objectives

- Understand how public, private, hybrid, and multi-cloud architectures differ.
- Learn the practical use cases and characteristics of each deployment type.
- Identify how organizations choose between cloud deployment strategies.
- Build decision-making clarity for workload and infrastructure planning.

Public Cloud: Key Concepts

- Cloud services delivered over shared infrastructure maintained by a provider.
- Users access compute, storage, and services through the provider’s environment.
- Infrastructure is not owned by the customer — it is rented as needed.
- Highly scalable and accessible from any connected location.
- Supports quick setup for new workloads and global accessibility.
- Ideal for startups, testing environments, and variable workloads.
- Managed entirely by the provider including updates and hardware.
- Used widely for analytics, app hosting, and distributed workloads.

Real-World Public Cloud Examples

- Amazon Web Services (AWS) — global cloud infrastructure.
- Microsoft Azure — extensive global network of data centers.
- Google Cloud Platform (GCP) — high-performance computing and data services.
- IBM Cloud — enterprise-focused cloud platform.
- Oracle Cloud Infrastructure — optimized for enterprise workloads.
- Alibaba Cloud — leading cloud provider in Asia.
- Tencent Cloud — public cloud services for digital applications.
- Used by companies needing large-scale computing without hardware ownership.

Private Cloud: Key Concepts

- Cloud infrastructure dedicated exclusively to one organization.
- Hosted either on-premises or in a dedicated off-site facility.
- Provides enhanced isolation, governance, and customization.
- Organization controls hardware, network, and security tools.
- Suitable for strict compliance, regulatory, or data-sensitive workloads.
- Offers predictable performance due to fully isolated resources.
- Requires skilled teams for setup, monitoring, and maintenance.
- Ideal for governments, financial institutions, and large enterprises.

Private Cloud Deployment Examples

- VMware-based private cloud environments used by enterprises.
- OpenStack private cloud deployments for IT flexibility.
- HPE GreenLake private cloud services.
- Dell APEX private cloud models.
- IBM Cloud Private — enterprise-grade private platform.
- Oracle Private Cloud Appliance for internal workloads.
- Cisco UCS-based private cloud deployments.
- Used for ERP systems, sensitive databases, and internal apps.

Hybrid Cloud: Key Concepts

- Combines private and public cloud environments into one architecture.
- Workloads can move between clouds based on business needs.
- Enables balancing sensitive data on private cloud and scalable workloads on public cloud.
- Allows gradual migration instead of full cloud transformation.
- Supports disaster recovery using public cloud as backup.
- Requires secure connectivity between environments.
- Offers flexibility for application deployment strategies.
- Useful for organizations with both legacy and modern workloads.

Hybrid Cloud Usage Examples

- Running customer-facing apps on public cloud and databases on private cloud.
- Using public cloud for burst capacity during peak demand.
- Backup and disaster recovery hosted on public cloud services.
- Sensitive financial or medical data kept in private cloud.
- Development/testing in public cloud; production in private cloud.
- Gradual migration from datacenter to cloud over several phases.
- Large enterprises maintaining legacy systems while expanding cloud use.
- Ideal for hybrid IT strategies requiring control + scalability.

Multi-Cloud: Key Concepts

- Uses more than one public cloud provider simultaneously.
- Allows distributing workloads across AWS, Azure, GCP, etc.
- Prevents dependence on a single cloud vendor.
- Enables choosing best-fit services from different providers.
- Requires consistent management and monitoring across clouds.
- Offers resilience in case one provider faces downtime.
- Useful for global presence and latency optimization.
- Aligns with modern digital transformation strategies.

Multi-Cloud Usage Examples

- Hosting analytics on GCP and enterprise workloads on AWS.
- Using Azure AD for identity and AWS for compute.
- Running regional workloads on different providers for latency reasons.
- Distributing critical apps across clouds for high availability.
- Leveraging specialized AI/ML tools from specific providers.
- Using multiple clouds to meet regional data laws.
- Scaling globally by utilizing multi-provider infrastructure.
- Common in multinational enterprises and SaaS companies.

Deployment Selection Criteria

- **Security Requirements:** Sensitive workloads → private or hybrid.
- **Cost Considerations:** Public cloud favored for unpredictable workloads.
- **Compliance & Regulations:** Industries like healthcare/banking → private or hybrid.
- **Performance Needs:** Low-latency apps may require hybrid or private deployment.
- **Workload Type:** Legacy systems → private; modern apps → public/hybrid.
- **Scalability Needs:** Public or multi-cloud suits fluctuating demand.
- **Vendor Strategy:** Multi-cloud for avoiding vendor lock-in.
- **IT Skillset:** Private cloud requires in-house expertise; public cloud lowers operational load.

Summary of the Lecture

- Public, private, hybrid, and multi-cloud offer distinct deployment approaches.
- Public cloud provides accessible, scalable services on shared infrastructure.
- Private cloud offers dedicated control for sensitive workloads.
- Hybrid cloud blends flexibility with security by mixing environments.
- Multi-cloud leverages multiple providers for resilience and specialized capabilities.
- Deployment selection depends on security, performance, cost, and compliance.
- Organizations often combine these models for optimized IT strategies.
- This PPT strictly covers only the assigned topics, excluding all others.

Next Lecture

NIST cloud reference architecture, layered components (consumer, provider, carrier, broker, auditor), architectural patterns

Quiz/ FAQ's

1. Can organizations use more than one deployment model?

Yes — hybrid and multi-cloud strategies combine multiple models effectively.

2. Is multi-cloud the same as hybrid cloud?

No — hybrid mixes private + public; multi-cloud mixes multiple public providers.

3. Why choose private cloud?

For strict governance, customization, and data sensitivity.

4. Why do companies adopt multi-cloud?

To access best-fit services and avoid vendor dependency.

5. Can hybrid cloud support legacy systems?

Yes — it lets legacy apps stay on private cloud while new apps run on public cloud.

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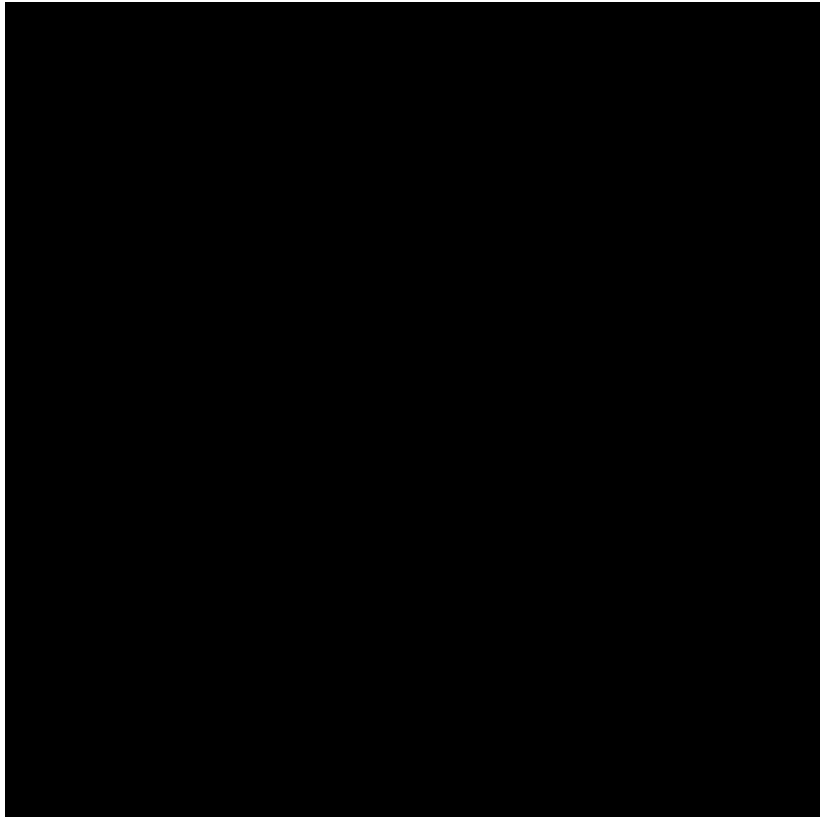
https://www.oracle.com/cloud/what-is-hybrid-cloud/
VMware — Multi-Cloud Architecture
https://www.vmware.com/topics/glossary/content/multi-cloud.html
Gartner — Multi-Cloud Strategy Insights
https://www.gartner.com/en/articles/what-is-multicloud
Red Hat — Hybrid Cloud Concepts
https://www.redhat.com/en/topics/cloud-computing/what-is-hybrid-cloud

Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- Cloud Computing: https://onlinecourses.nptel.ac.in/noc21_cs14/preview
- Cloud Computing: https://onlinecourses.nptel.ac.in/noc26_cs55/preview
- Google Cloud Computing Foundations: https://onlinecourses.nptel.ac.in/noc20_cs55/preview
- Google Cloud Computing Foundations: https://onlinecourses.nptel.ac.in/noc24_cs131/preview

Class-Wise Feedback



Thank You

For queries
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